

Hospital Patient & Operations Analytics

SQL-Based Data Analysis Project

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Tool	MySQL / SQL
Working table	hos_op
Patients analyzed	1,980
Business questions	25

Portfolio Project Report

1. Project Overview

This project analyzes hospital patient and operational data using SQL to identify patterns in patient volume, admissions, diagnoses, treatment costs, readmissions, treatment types and departmental performance. The analysis was structured around business questions so that SQL results could be translated into practical operational insights.

Objectives

- Understand patient distribution across departments and cities.
- Identify common diagnoses, admission types and treatment types.
- Measure treatment-cost patterns and departmental cost differences.
- Measure readmission volume and readmission rates.
- Apply advanced SQL techniques for comparisons and ranking.
- Translate findings into business-oriented recommendations.

Scope

The working table contained 1,980 rows at the time of analysis. The project is descriptive analytics and does not attempt to establish medical causation.

2. Dataset & Data Preparation

The dataset contains patient-level information covering demographics, admission details, department, diagnosis, treatment type, length of stay, treatment cost, insurance, payment method, readmission status and patient rating.

Field	Purpose
Patient_ID	Patient identifier
Age / Gender / City	Patient demographics
Admission_Date / Discharge_Date	Hospital stay dates
Department / Doctor	Operational assignment
Diagnosis / Treatment_Type	Recorded diagnosis and treatment categories
Admission_Type	Admission classification
Length_of_Stay	Hospital stay duration
Treatment_Cost	Recorded treatment cost
Insurance / Payment_Method	Coverage and payment information
Readmitted / Patient_Rating	Readmission status and patient feedback

Data preparation

- **Duplicate check:** Patient_ID duplicates were checked in the working table; no duplicate Patient_ID values were found.
- **NULL awareness:** Selected fields contained NULL values. COUNT(column) was treated carefully because it excludes NULL values.
- **Treatment cost:** Treatment_Cost was imported as text because of currency formatting.
- **Numeric conversion:** REPLACE() removed the imported currency marker and commas, and CAST(... AS DECIMAL) enabled numeric aggregation.

3. Analytical Approach & SQL Techniques

The workflow followed a client-style process: business requirement → business question → SQL analysis → finding → business interpretation.

Technique	Application
COUNT / SUM / AVG	Patient counts, treatment costs, average stay and average cost.
GROUP BY	Department, city, diagnosis, admission and treatment comparisons.
CASE WHEN	Conditional readmission counts and age-group classification.
HAVING	Filtering grouped departments by a readmission-rate condition.
Subquery	Comparing department readmission rates with the hospital-wide rate.
CTE	Breaking the treatment-cost ranking into readable stages.
ROW_NUMBER()	Ranking treatments within each department.

Advanced analysis pattern

The final analysis first calculated average treatment cost for each Department + Treatment_Type combination. A window function then ranked treatments within each department, allowing the top-cost treatment in every department to be selected.

4. Key Findings

The following results were recorded from the completed analysis.

Area	Finding
Patient volume	1,980 patients analyzed.
Gender distribution	Male = 990; Female = 990.
Top city	Patna — 229 patients.
Top department by patient volume	Cardiology — 275 patients.
Most common diagnosis	Heart Disease — 212 patients.
Average patient age	43.07 years.
Most common admission type	Emergency — 689 admissions.
Total treatment cost	■214,608,018.
Highest departmental total cost	ENT — ■29,840,709.
Highest departmental average cost	Cardiology — ■11,345.92.
Readmitted patients	996 patients; overall readmission rate = 50.30%.
Highest readmitted patient count	Neurology — 141 patients.
Average stay comparison	Readmitted = 7.9 days; non-readmitted = 8.1 days.
Highest average patient rating	General Medicine.
Most common treatment type	Surgery — 296 patients.
Insurance marked Yes	1,008 patients.
Most-used payment method	Card — 519 patients.
Highest average treatment-type cost	Radiation.
Highest-cost department/treatment result	General Medicine → Chemotherapy — ■157,339 average treatment cost.

5. Business Interpretation

The findings point to several areas that hospital management could investigate further. These are operational interpretations, not medical conclusions.

High readmission areas

Neurology recorded the highest number of readmitted patients. Departments whose readmission rate is above the hospital-wide rate can be prioritized for deeper analysis of diagnosis, admission type, treatment type and length-of-stay patterns.

High-cost areas

ENT had the highest total treatment cost, while Cardiology had the highest average departmental treatment cost. Breaking these totals down by treatment type and diagnosis can help identify cost concentration.

Length of stay and cost

General Medicine showed the highest average length of stay in the completed comparison and also ranked highest on average treatment cost in that analysis. A deeper review of case mix and treatment categories could help explain this pattern.

Treatment-cost concentration

Radiation had the highest average treatment cost among treatment types, while the department-level ranking identified Chemotherapy as the highest-cost treatment for General Medicine.

6. Business Recommendations

Recommendation	Suggested action
Prioritize readmission investigation	Review departments above the hospital-wide readmission rate and identify common diagnosis, admission and treatment patterns.
Analyze cost drivers	Break high-cost departments down by treatment type and diagnosis to identify major cost concentrations.
Monitor long-stay departments	Track higher average length of stay alongside treatment cost and admission type.
Build department-level monitoring	Create a dashboard combining patient volume, average cost, total cost, average stay, readmission rate and patient rating.
Review expensive treatments	Compare treatment cost with treatment frequency so high average cost is not confused with high utilization.
Create repeatable reporting	Reuse the SQL queries with updated data for recurring operational reporting.

7. Conclusion

This project demonstrates an end-to-end SQL analytics workflow on hospital operations data. The analysis moved from basic patient and department profiling to cost analysis, readmission analysis and advanced ranking problems.

The portfolio value of the project comes from combining technical SQL skills with business interpretation. The analysis did not stop at producing numbers; it translated results into areas that hospital management could investigate further.

The project includes aggregation, conditional logic, data-type cleaning, subqueries, CTEs and window functions. The complete SQL query set is maintained separately in the project SQL file.

Portfolio takeaway

This project can be presented as a SQL-based Healthcare Analytics project demonstrating data cleaning, exploratory analysis, advanced SQL and business-oriented insight generation.

Selected SQL patterns

Pattern	SQL
Readmission count	<code>SUM(CASE WHEN Readmitted = 'Yes' THEN 1 ELSE 0 END)</code>
Readmission rate	<code>SUM(CASE WHEN Readmitted = 'Yes' THEN 1 ELSE 0 END) / COUNT(*) * 100</code>
Cost conversion	<code>CAST(REPLACE(REPLACE(Treatment_Cost, '?', ''), ',', '') AS DECIMAL(10,2))</code>
Within-department ranking	<code>ROW_NUMBER() OVER (PARTITION BY Department ORDER BY avg_treat_cost DESC)</code>

End of report